CLOUD ASSIGNMENT

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1. Research on Microsoft Active Directory vs Microsoft Entra ID

Introduction:-

Microsoft provides identity and access management solutions for organizations. Two important solutions are Microsoft Active Directory (AD DS) and Microsoft Entra ID. Active Directory is mainly used for traditional on-premises networks, while Entra ID is a cloud-based identity service.

1.Microsoft Active Directory

Active Directory Domain Services (AD DS) is a directory service used to manage users, computers, groups, and other resources in an organization's network.

Main features:

• Manages users and computers
•Provides authentication and access control
•Supports Group Policy
•Uses Kerberos and LDAP
•Mainly works with on-premises Windows networks

2.Microsoft Entra ID

Microsoft Entra ID is a cloud-based identity and access management service. It is commonly used with Microsoft 365, Azure, and cloud applications.

Main feature:

Cloud-based user management
Single Sign-On (SSO)
Multifactor Authentication (MFA)
Conditional Access
Passwordless authentication
Access management for cloud applications

Hybrid Identity:
Organizations can use both Active Directory and Microsoft Entra ID. User identities from on-premises Active Directory can be synchronized with Entra ID, allowing users to access both local and cloud services.

Conclusion:
Active Directory is mainly designed for on-premises network management, while Microsoft Entra ID is designed for cloud-based identity and access management. Both can work together in a hybrid environment.

2.Basics of Linux Commands

Linux commands are instructions used to perform different tasks through the Terminal. They are used for managing files, directories, users, permissions, and system information.

Common Linux Commands:

Command	Use	Example
pwd	Show current directory	pwd
ls	List files & folders	ls
cd	Change directory	cd Documents
mkdir	Create directory	mkdir project
rmdir	Remove empty directory	rmdir project
touch	Create file	touch file.txt
cat	Display file contents	cat file.txt
cp	Copy file/folder	cp a.txt b.txt
mv	Move/rename file	mv old.txt new.txt
rm	Delete file	rm file.txt
clear	Clear terminal	clear
whoami	Show current user	whoami
date	Show date & time	date
man	Show command help	

File and Directory Example

mkdir college

cd college

touch notes.txt

ls

cat notes.txt

cd ..

These commands create a folder, enter it, create a file, list its contents, display the file, and return to the previous directory.

3.Types of Identity

Introduction:

Identity is the information used to identify and verify a person, device, application, or service in a computer system. Identity management helps ensure that only authorized users and systems can access resources

Types of Identity:

Type	Description
User Identity	Identifies a person using a system or application.
Device Identity	Identifies a computer, mobile phone, or other device.
Application Identity	Identifies an application or software.
Service Identity	Identifies a service or system process.
Guest Identity	Identifies external users who are given limited access.
Machine Identity	Identifies machines and enables secure communication between systems.

Importance :

Provides secure access to resources.

Prevents unauthorized access.

Helps manage users and devices efficiently.

Supports authentication and authorization.

Protects sensitive information and improves security.

Helps organizations track and control access.

Conclusion :

Identity is an important part of cybersecurity and access management. Different types of identities help organizations securely manage users, devices, applications, and services while preventing unauthorized access.

4.Difference Between User, Group and Role

User :

Represents an individual person or account.

Has a unique identity.

Example: Abhina

Used to identify and authenticate a person.

A user can belong to multiple groups

Group:

Collection of users with common requirements.

Contains multiple users.

Example: BCA Students

Used to manage users together.

A group can contain many users.

Role:

Defines what permissions or actions a user can perform.

Contains a set of permissions.

Example: Administrator

Used to control access to resources.

A user can have one or more roles.

In simple words:

User = Who you are

Group = Who you are with

Role = What you are allowed to do

conclusion:

Users, groups, and roles are important for identity and access management. A user represents an individual, a group organizes users, and a role defines their permissions. Together, they help organizations manage access securely and efficiently.

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